

Economic Complexity Homework 3 Research Proposal

Research topic

The exposure of supply chains to exogenous shocks in highly complex economies.

Relevance

While economic complexity has enabled countries to specialize in high-value, knowledge-intensive industries, it has also increased their dependence on globally distributed networks of inputs that behave like small-world systems. Due to the recent unpredictable economic environment and continuously appearing challenges (financial crisis, covid pandemic, Russo-Ukrainian war etc.) there has been a growing importance of the question regarding the fragility of these global supply chains and how vulnerable production systems are.

Related research

Based on Hausman's work, it has been shown that developed countries that specialize in high complexity products are usually relying on a large set of complementary, often globally produced capabilities (Hidalgo & Hausmann, 2009). In 2012 it has been discovered that modern economies are highly interconnected, and because production networks have small-world structures, micro-level shocks can scale up and lead to large macroeconomic fluctuations (Acemoglu, Carvalho, Ozdaglar, & Tahbaz-Salehi, 2012). This was also shown in 2019 by Inoue and Todo, who provided firm-level evidence on that supply chains display small-world characteristics, furthermore that even a local shock can affect not directly involved parties through short network paths (Inoue & Todo, 2019).

Research question, possible stakeholders

As above introduced, there are even more research on production networks and how those amplify shocks, however there is still little empirical evidence on **how a country's level of economic complexity interacts with and how it shapes its supply-chain network's resilience to disruptions**. The findings of such analysis could benefit not only policymakers but also influence firms' decisions who are relying on global input sourcing.